

Varenya Chivukula



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SUMMARY

Final-year Computer and Autonomous Systems Engineering student with industry experience in industrial digitalization, AI/computer vision, embedded systems and software development. Builds practical engineering solutions across autonomous robotics, IoT and data-driven systems; active in research, technical leadership and interdisciplinary projects.

EDUCATION

University of Wollongong in Dubai (UOWD) | Dubai, UAE

Bachelor of Engineering (Honours) in Computer and Autonomous Systems Engineering | Sept 2023 – Present | Grade: HD (High Distinction)

PROFESSIONAL EXPERIENCE

SpacePoint | UAE | Facilitator / Engineering Intern | Jan 2026 – Present

- Developing an attitude determination and control systems (ADCS) PCB schematic and layout in KiCad for CubeSat engineering work; hardware validation planned.
- Deliver satellite engineering and hardware workshops; helped coordinate a proposed SpacePoint–AIESEC collaboration.

ADNOC Group | Abu Dhabi, UAE | Digital Technical Solutions Intern | Jul 2025 – Sep 2025

- Developed a real-time computer-vision safety monitoring prototype using YOLOv8, OpenCV, FastAPI and React, with detection workflows and structured event logging.
- Integrated scene classification and a locally hosted LLM to produce contextual summaries of safety events.
- Built AVEVA PI Vision dashboards for industrial operations and audited 500+ PI System tags to support reliable asset data visibility

TECHNICAL PROJECTS

AVAA – Intelligent Child Safety and Interaction Robot | Sept 2025 – Jul 2026

National Runner-Up — James Dyson Award, UAE National Stage 2026

- Developed a privacy-focused autonomous companion robot integrating on-device YOLOv8 perception, child following, hazard monitoring and sensor-based obstacle avoidance.
- Integrated Raspberry Pi-based processing, ROS 2 robotics and offline speech interaction into a multidisciplinary working prototype

SILAHGRID Distributed Edge AI Infrastructure Simulation | Aug 2025 – Sep 2025

- Modeled modular edge-computing pods with Python/SimPy and MATLAB, simulating workload distribution, solar energy, thermal behavior and adaptive task routing.

PowerSprout IoT System | Jan 2025 – July 2025

- Built an ESP32-enabled waste-to-energy prototype with sensor acquisition and a web dashboard for real-time electrical output monitoring.

RESEARCH & PUBLICATIONS

Adaptive Wireless Communication Management Framework for High-Density Multimedia Production Environments | *IEEE AIRoSIP 2026* | Co-author

Simulated adaptive resource allocation for dense wireless production networks; compared congestion, latency and throughput against static allocation.

Comparative Analysis of Classical and Quantum Vibration Sensors Under Radiation-Induced Environmental Stress | *IEEE EECSI 2026* | Co-author

Compared sensing performance and robustness of classical and quantum vibration sensing approaches in radiation-stressed conditions.

Comparative Study of RF Shielding Using SAR Simulations and Nanocomposite Materials | *9th ICEEIE UM 2025* | Co-author | *Best Paper Award*

Evaluated MXene, graphene and conductive composites using SAR simulations to study RF exposure reduction

TECHNICAL SKILLS

- Programming: Python, C, JavaScript, SQL, Bash | AI/Data: PyTorch, TensorFlow, YOLOv8, OpenCV, scikit-learn, Pandas
- Engineering: ROS 2, ESP32, MATLAB/Simulink, KiCad, Webots, AVEVA PI Vision | Software: FastAPI, Flask, React, Node.js, PostgreSQL, Git, Linux

LEADERSHIP & RECOGNITION

- Secretary, Engineers Australia Student Society (UOWD), 2025-Present - industry outreach, events and chapter operations.
- Deputy Head of Events, UOWD Tech Club, Apr - Oct 2025 - event planning and team coordination.
- Additional training: Samsung Innovation Campus AI Course (2024); Full-Stack Web Development Bootcamp (2024 - 2025).
- National Runner-Up — James Dyson Award, UAE National Stage 2026 — AVAA
- Best Paper Award — 9th ICEEIE UM Conference 2025 — RF Shielding Research